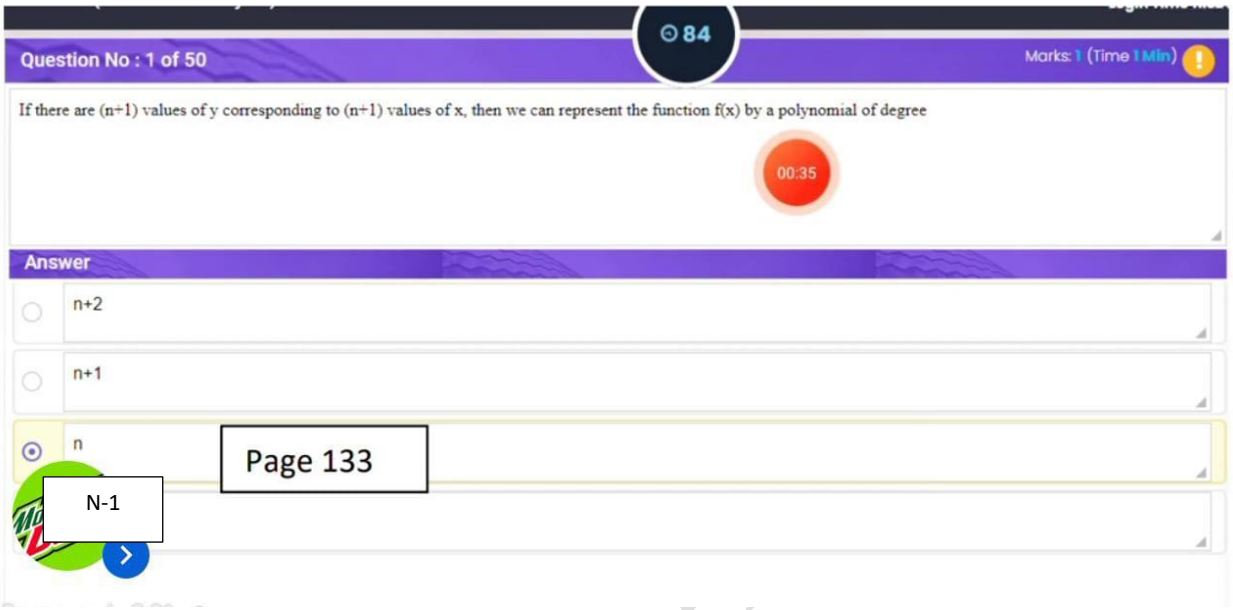
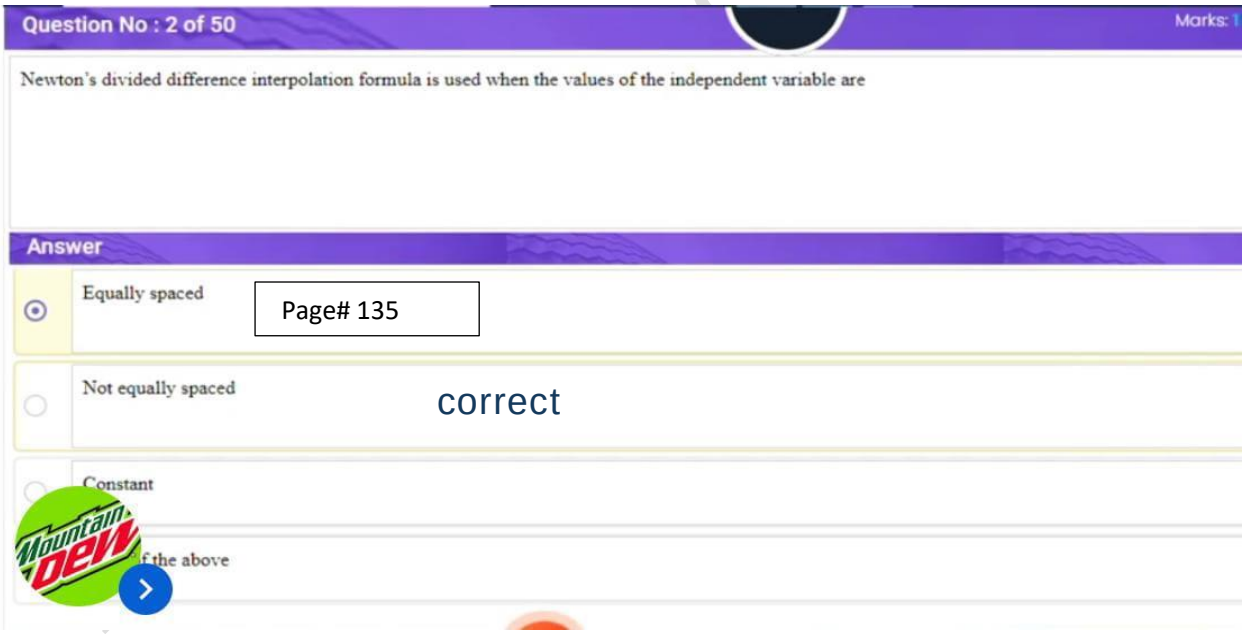
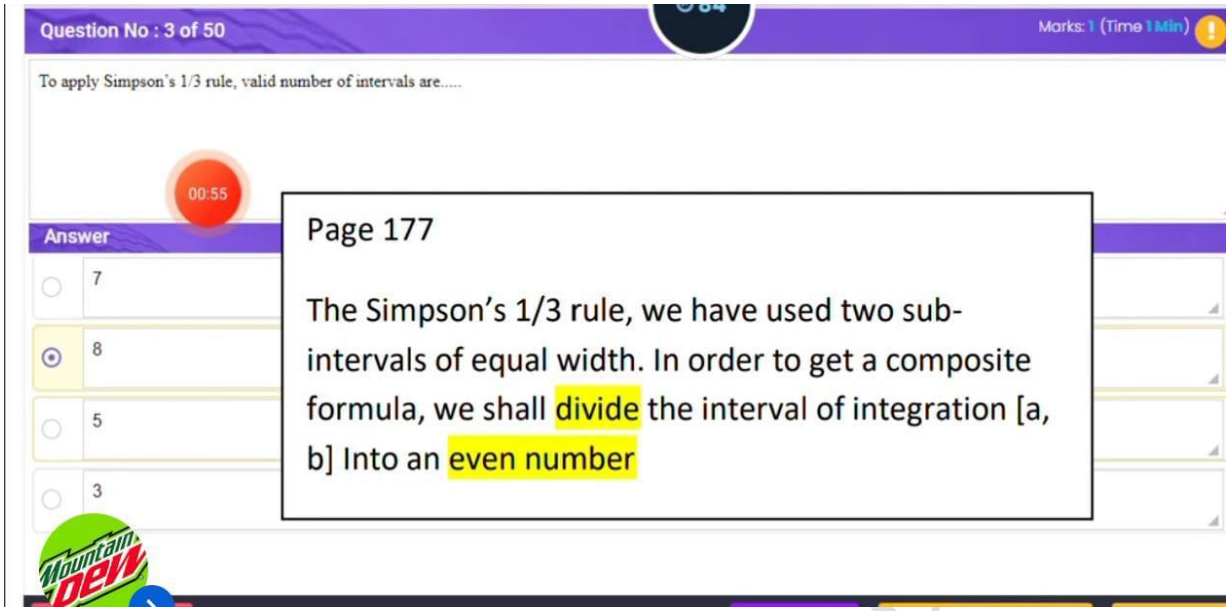
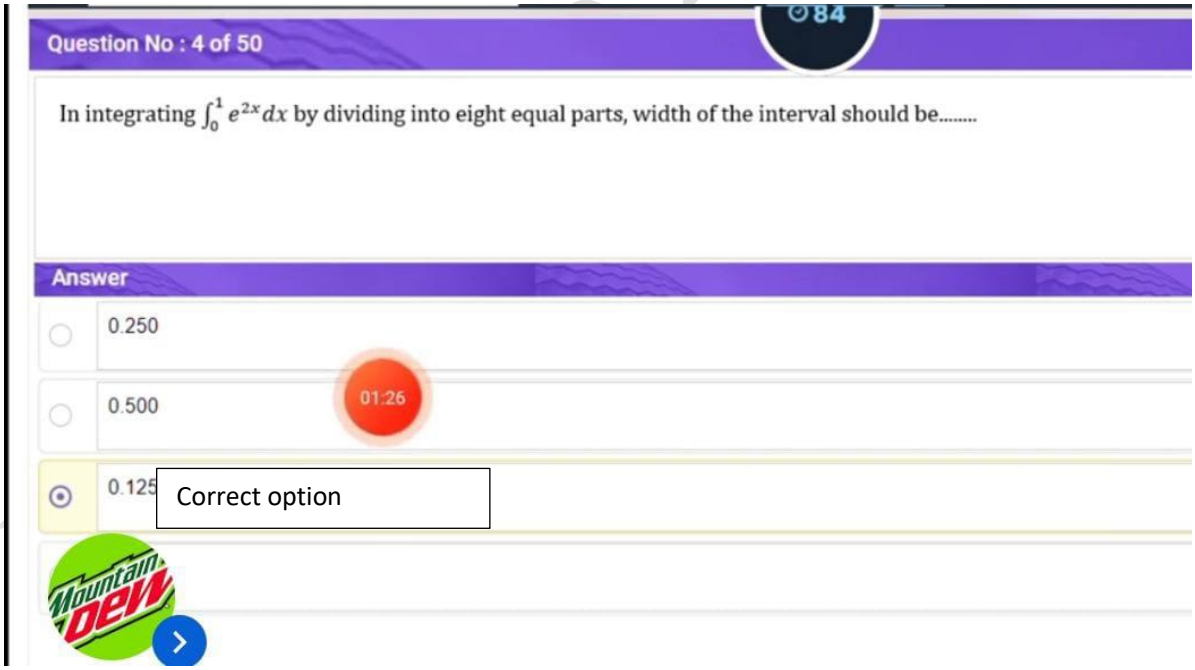
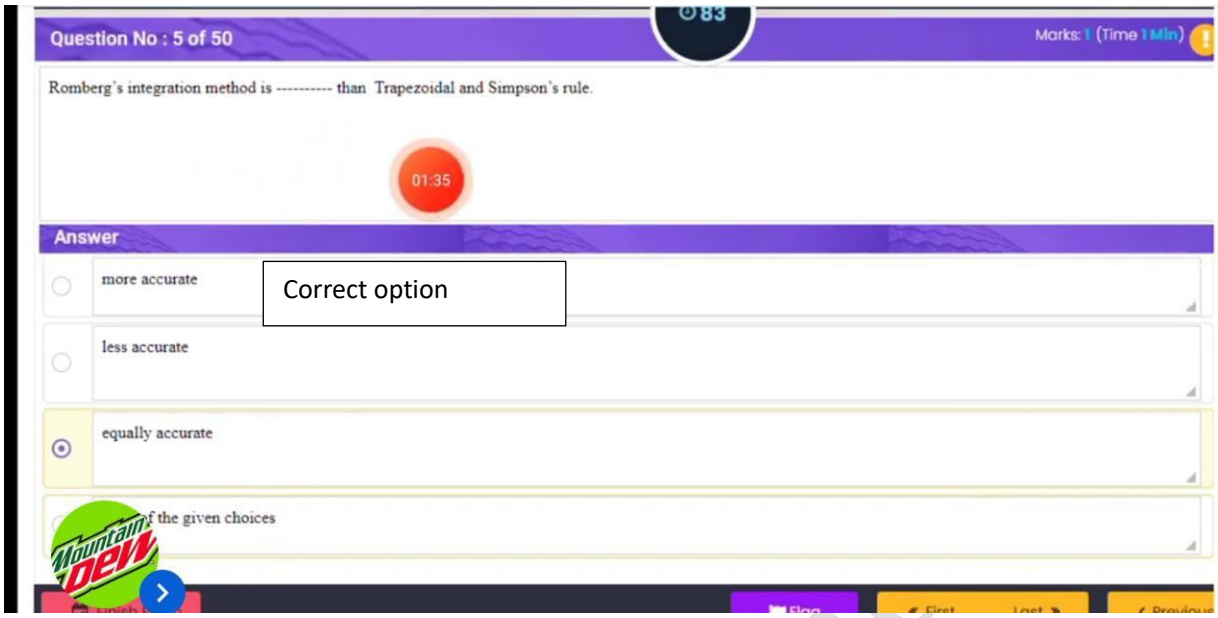
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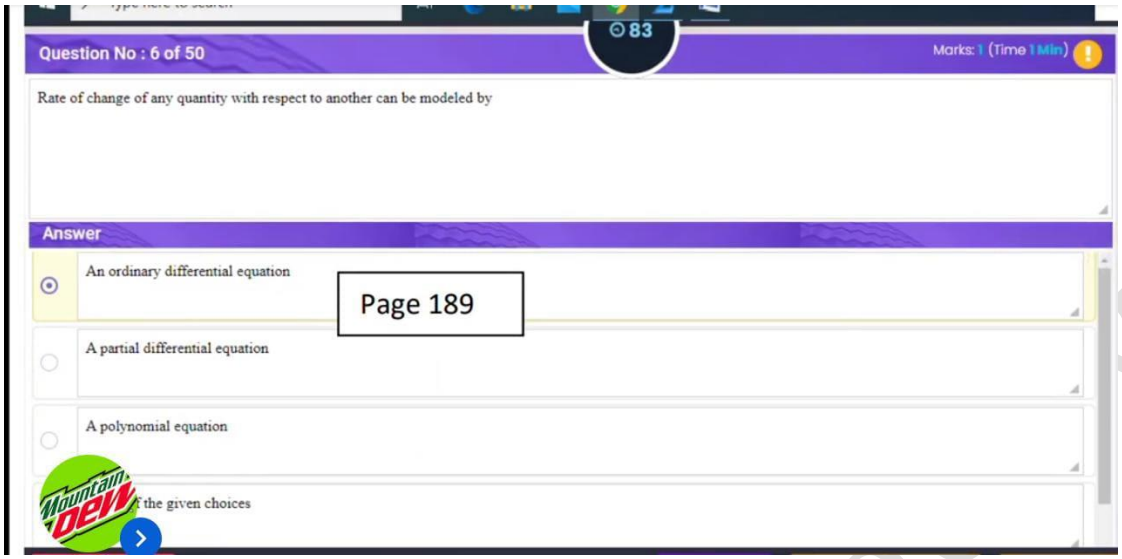


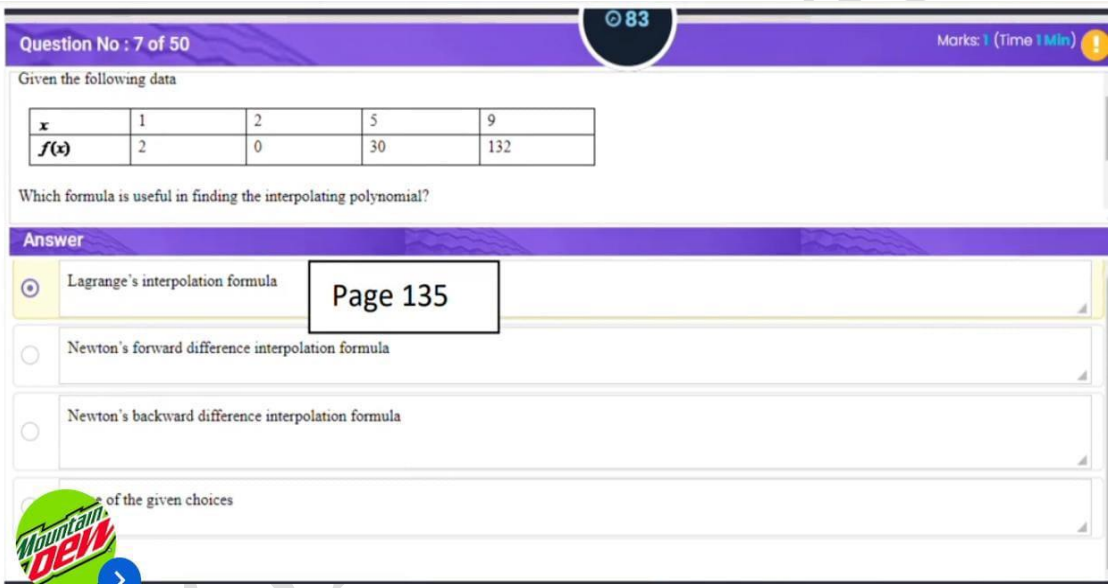

Question No : 3 of 50
Marks: 1 (Time 1 Min)
To apply Simpson's 1/3 rule, valid number of intervals are.....
00:55
Answer
7
8
5
3
Page 177
The Simpson's 1/3 rule, we have used two sub-intervals of equal width. In order to get a composite formula, we shall divide the interval of integration [a, b] Into an even number


Question No : 4 of 50
In integrating $\int_0^1 e^{2x}dx$ by dividing into eight equal parts, width of the interval should be.....
Answer
0.250
0.500
01:26
0.125
Correct option

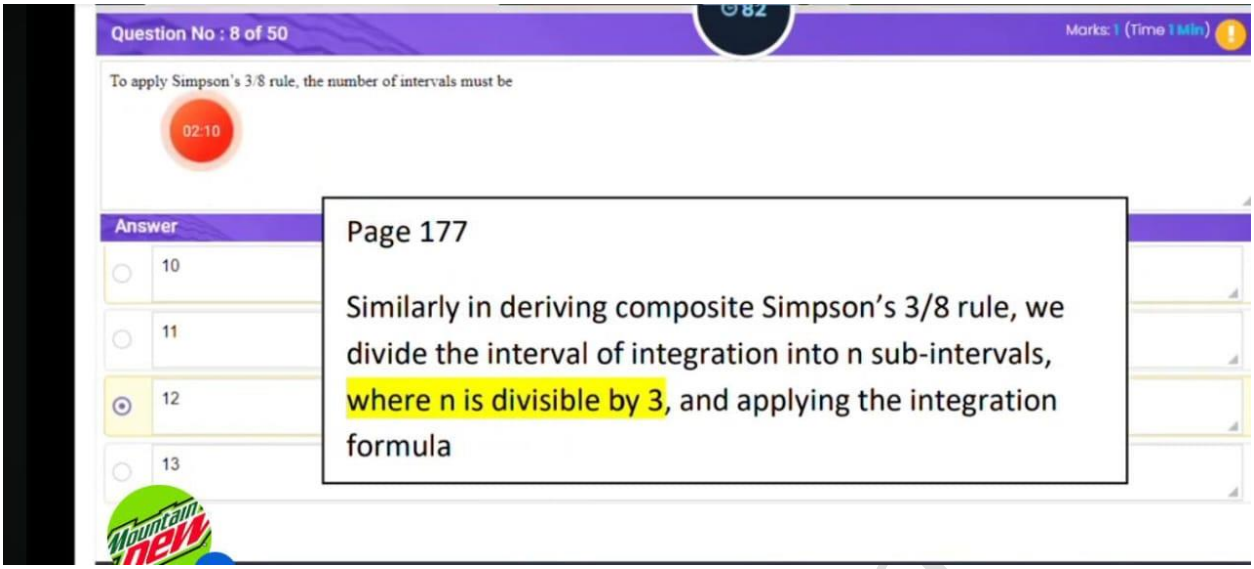


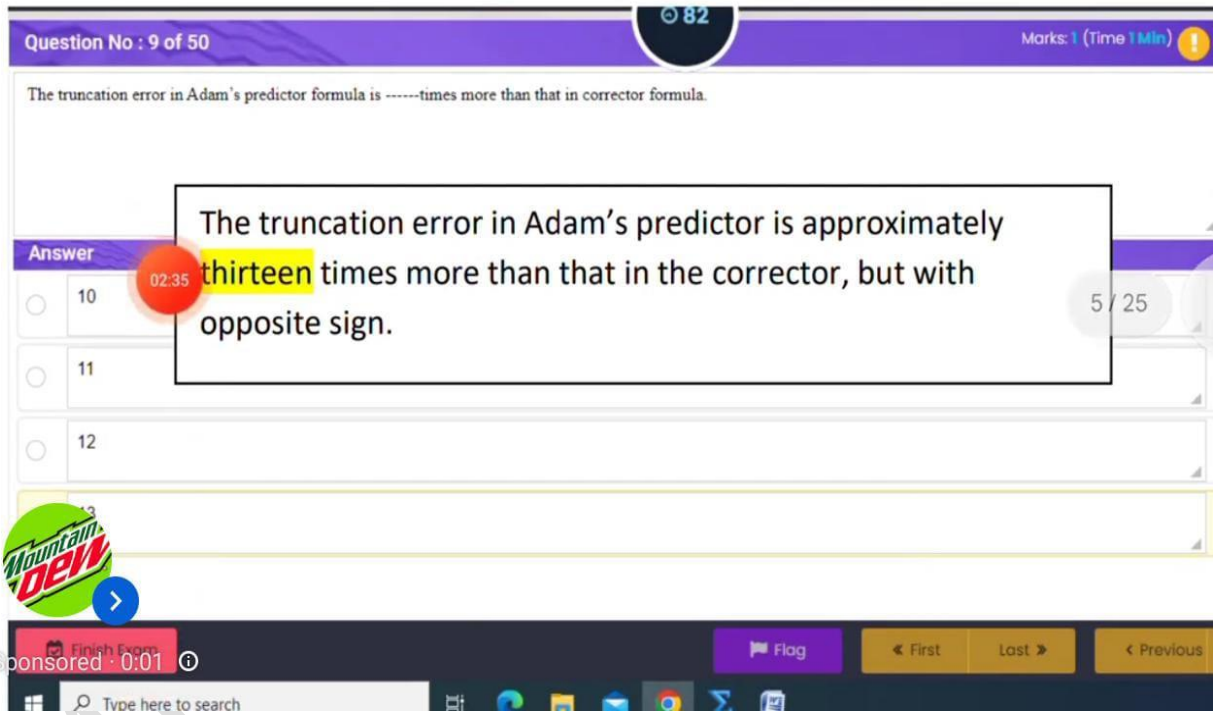
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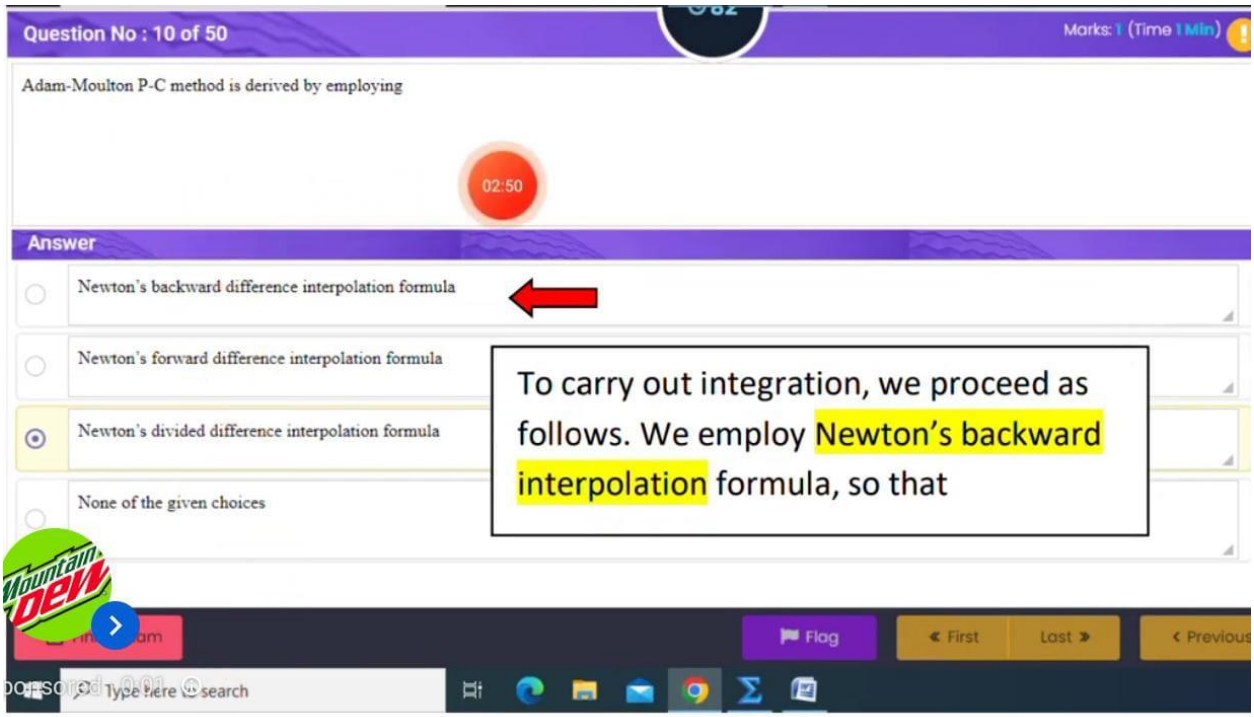

83
Question No : 6 of 50
Marks: 1 (Time 1 Min)
Rate of change of any quantity with respect to another can be modeled by
Answer
An ordinary differential equation
Page 189
A partial differential equation
A polynomial equation
the given choices

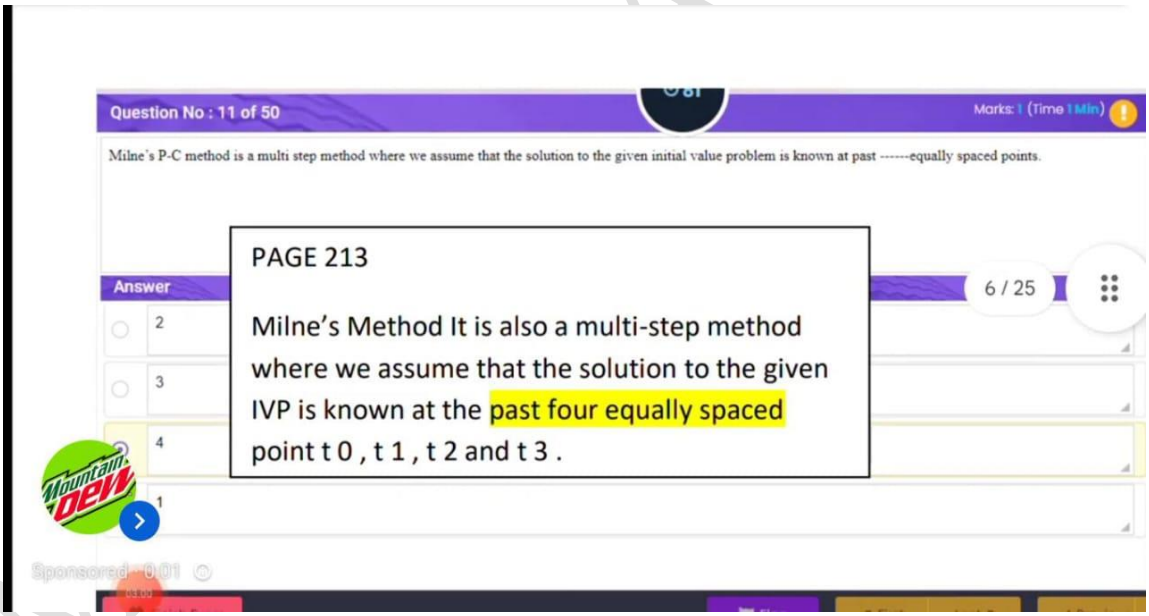

83
Question No : 7 of 50
Marks: 1 (Time 1 Min)
Given the following data
x 1 2 5 9
f(x) 2 0 30 132
Which formula is useful in finding the interpolating polynomial?
Answer
Lagrange's interpolation formula
Page 135
Newton's forward difference interpolation formula
Newton's backward difference interpolation formula
of the given choices

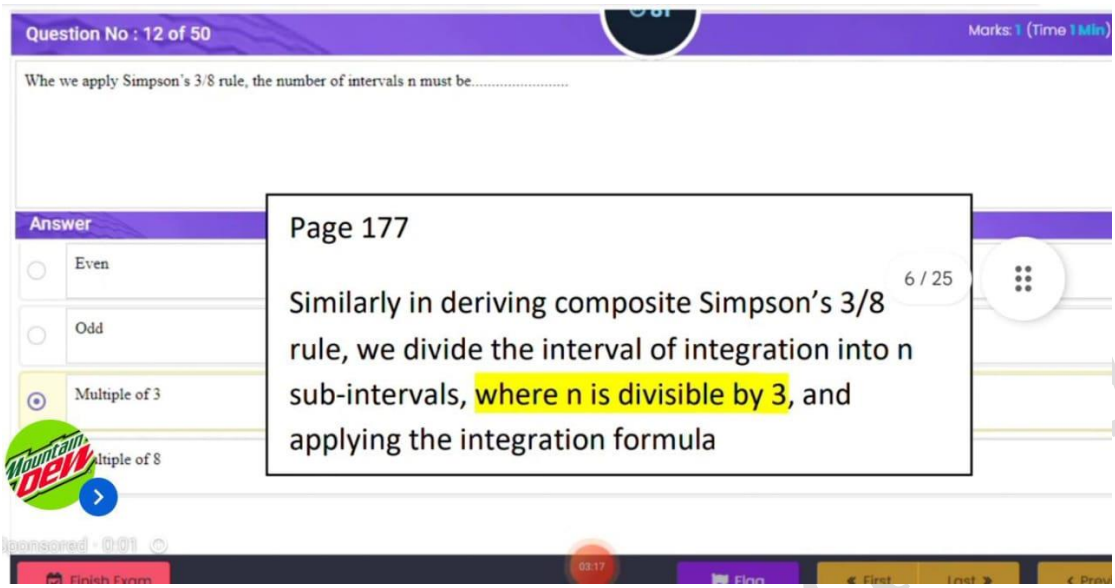
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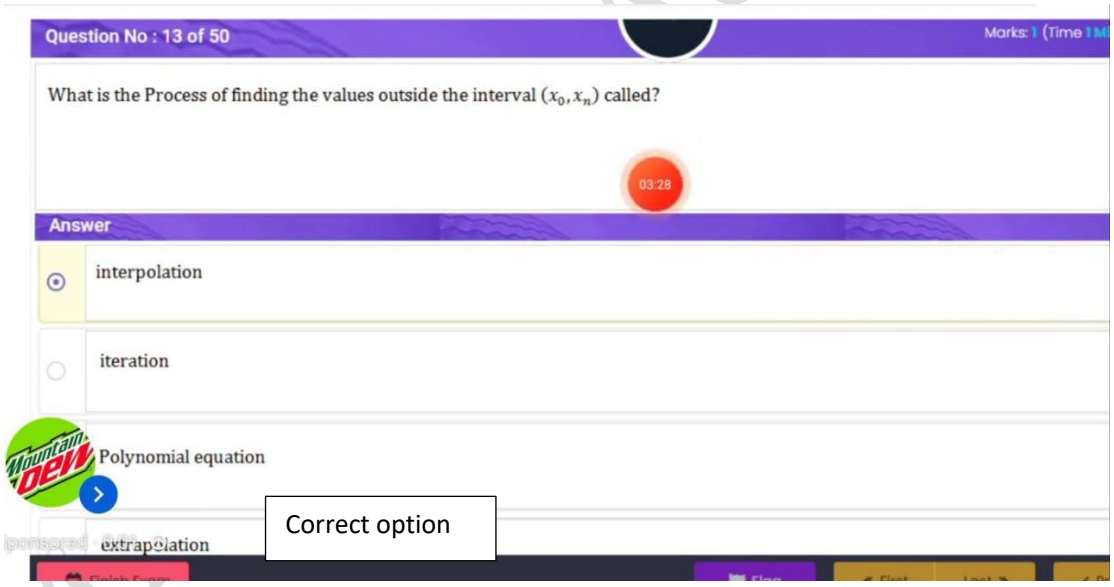

Question No : 8 of 50
Marks: 1 (Time 1 Min)
To apply Simpson's 3/8 rule, the number of intervals must be
02:10
Answer
10
11
12
13
Page 177
Similarly in deriving composite Simpson's 3/8 rule, we divide the interval of integration into n sub-intervals, where n is divisible by 3, and applying the integration formula

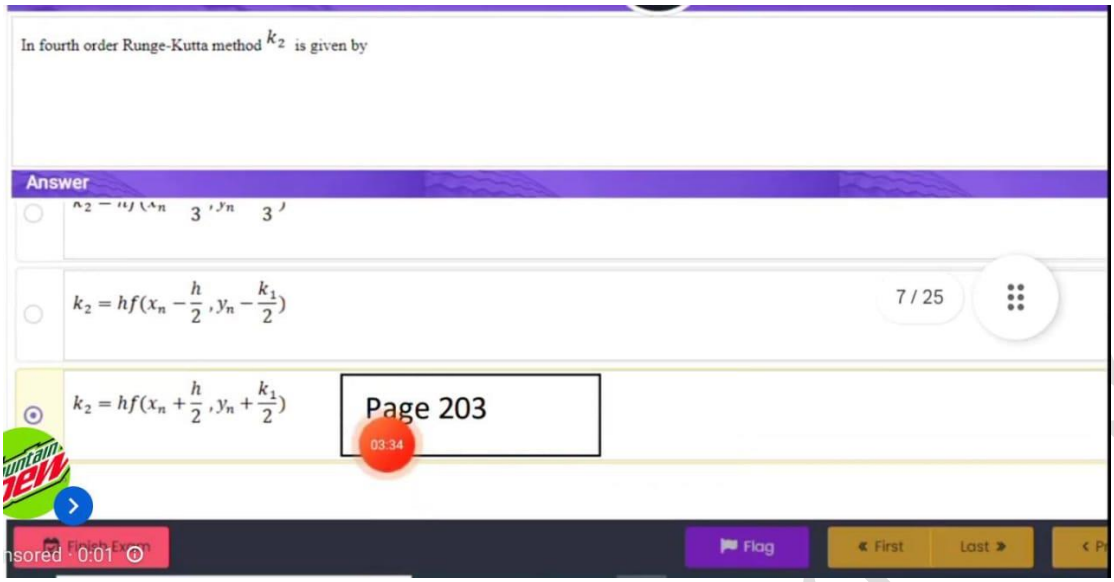

Question No : 9 of 50
Marks: 1 (Time 1 Min)
The truncation error in Adam's predictor formula is ------times more than that in corrector formula.
Answer
02:35
10
11
12
The truncation error in Adam's predictor is approximately thirteen times more than that in the corrector, but with opposite sign.
5 / 25
Finish Exam
Flag
« First
Last »
‹ Previous
Type here to search

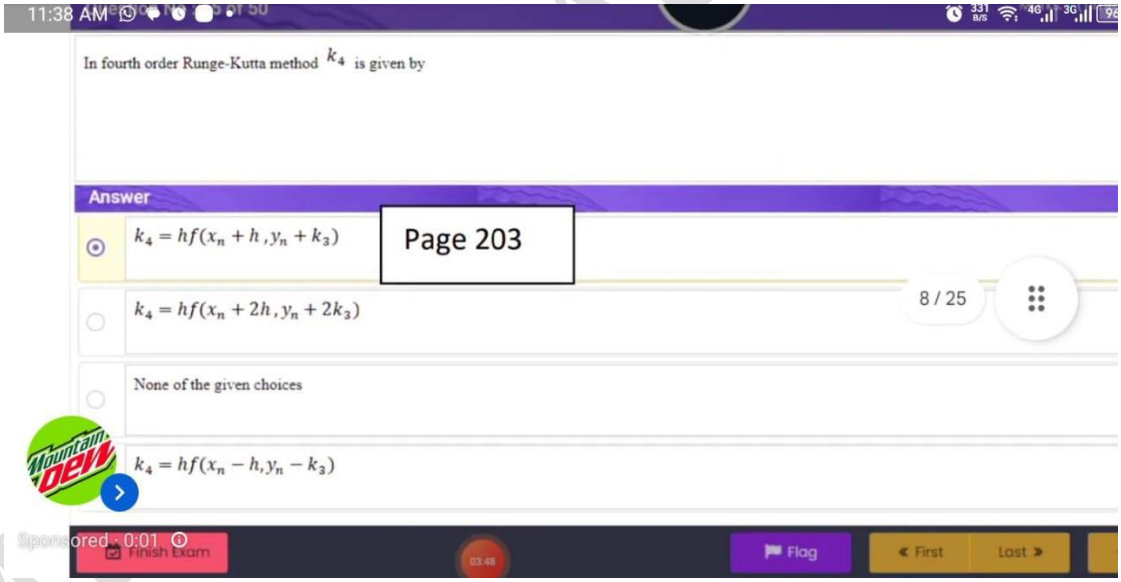

Question No : 10 of 50
Marks: 1 (Time 1 Min)
Adam-Moulton P-C method is derived by employing
02:50
Answer
Newton's backward difference interpolation formula
Newton's forward difference interpolation formula
Newton's divided difference interpolation formula
None of the given choices
To carry out integration, we proceed as follows. We employ Newton's backward interpolation formula, so that
Flag
« First
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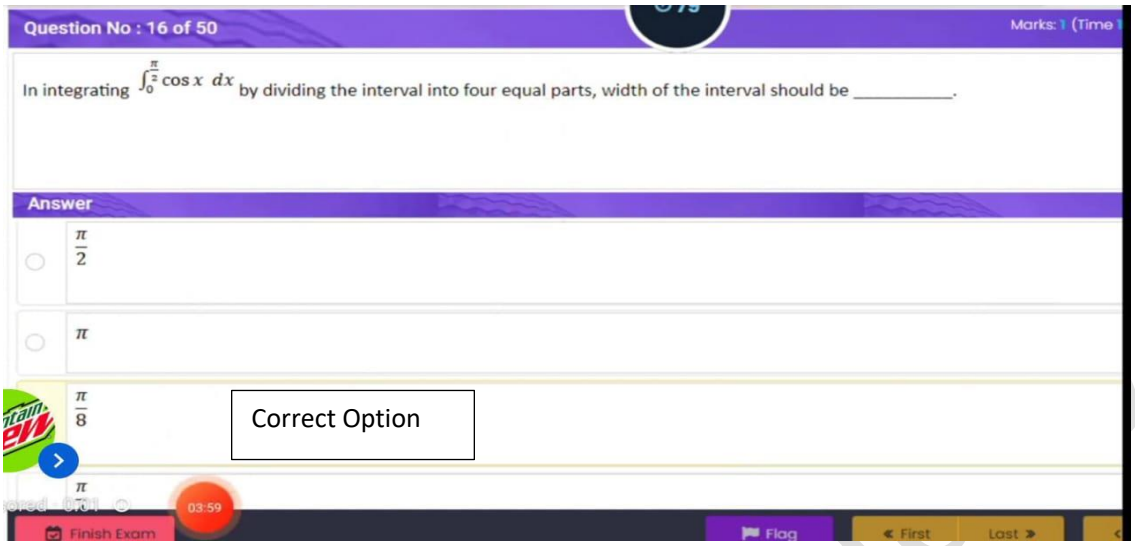

Question No : 11 of 50
Marks: 1 (Time 1 Min)
Milne's P-C method is a multi step method where we assume that the solution to the given initial value problem is known at past ------equally spaced points.
PAGE 213
Milne's Method It is also a multi-step method where we assume that the solution to the given IVP is known at the past four equally spaced point t 0 , t 1 , t 2 and t 3 .
Answer
6 / 25
2
3
4
1

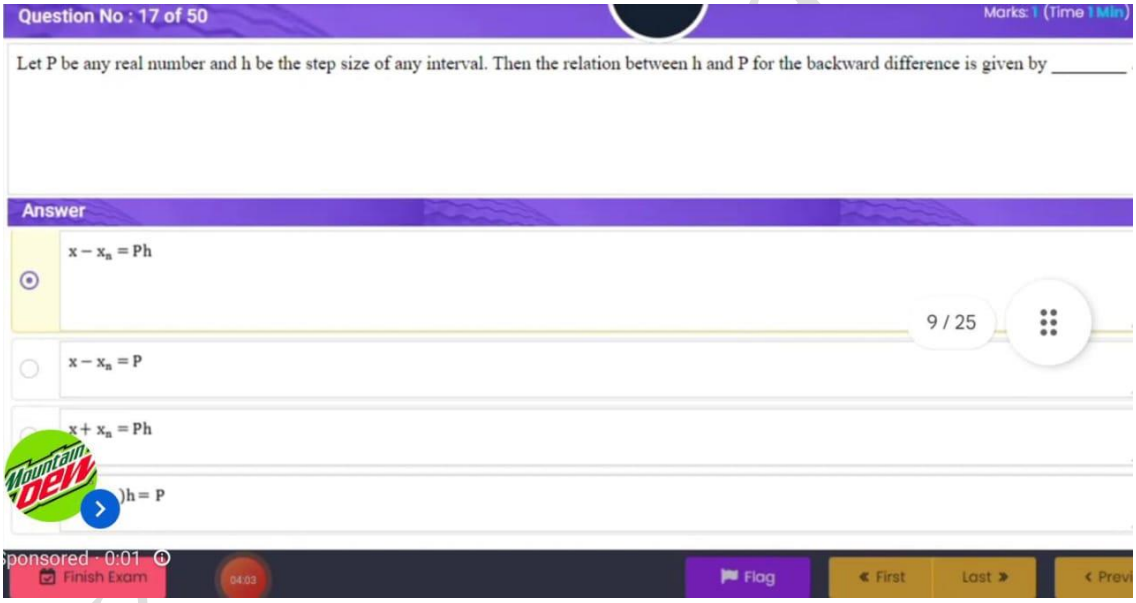

Question No : 12 of 50
Marks: 1 (Time 1 Min)
Whe we apply Simpson's 3/8 rule, the number of intervals n must be.....
Answer
Even
Odd
Multiple of 3
ltiple of 8
Page 177
6 / 25
Similarly in deriving composite Simpson's 3/8 rule, we divide the interval of integration into n sub-intervals, where n is divisible by 3, and applying the integration formula


Question No : 13 of 50
Marks: 1 (Time 1 M
What is the Process of finding the values outside the interval (x_0, x_n) called?
03:28
Answer
interpolation
iteration
Polynomial equation
extrapolation
Correct option

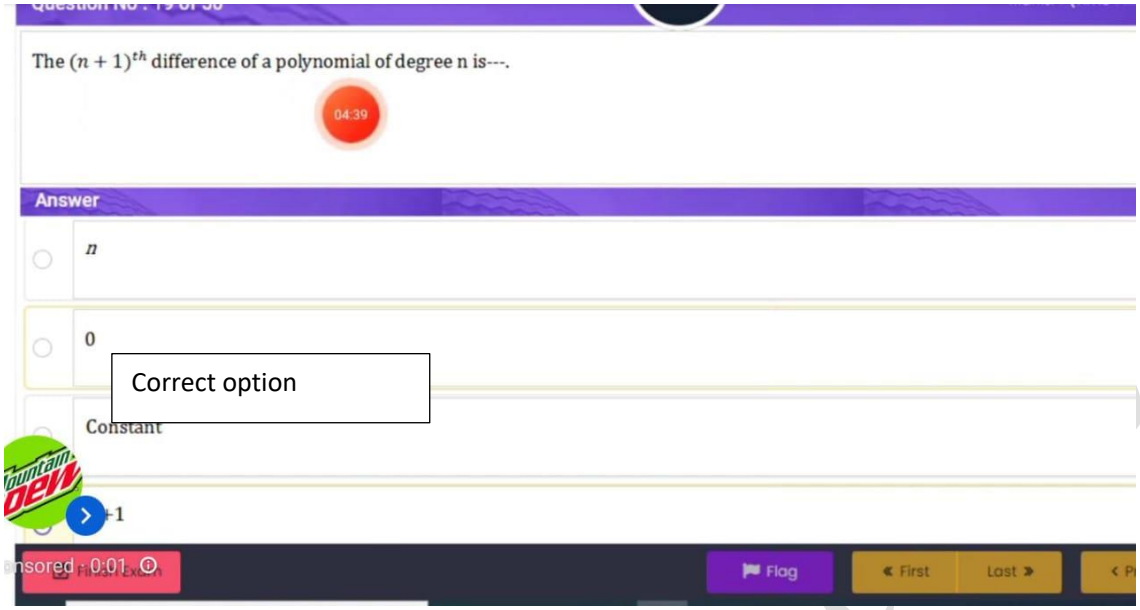

In fourth order Runge-Kutta method k_2 is given by
Answer
$k_2 = hf(x_n - \frac{h}{2}, y_n - \frac{k_1}{2})$
7 / 25
$k_2 = hf(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
Page 203
03:34
Flag
« First
Last »

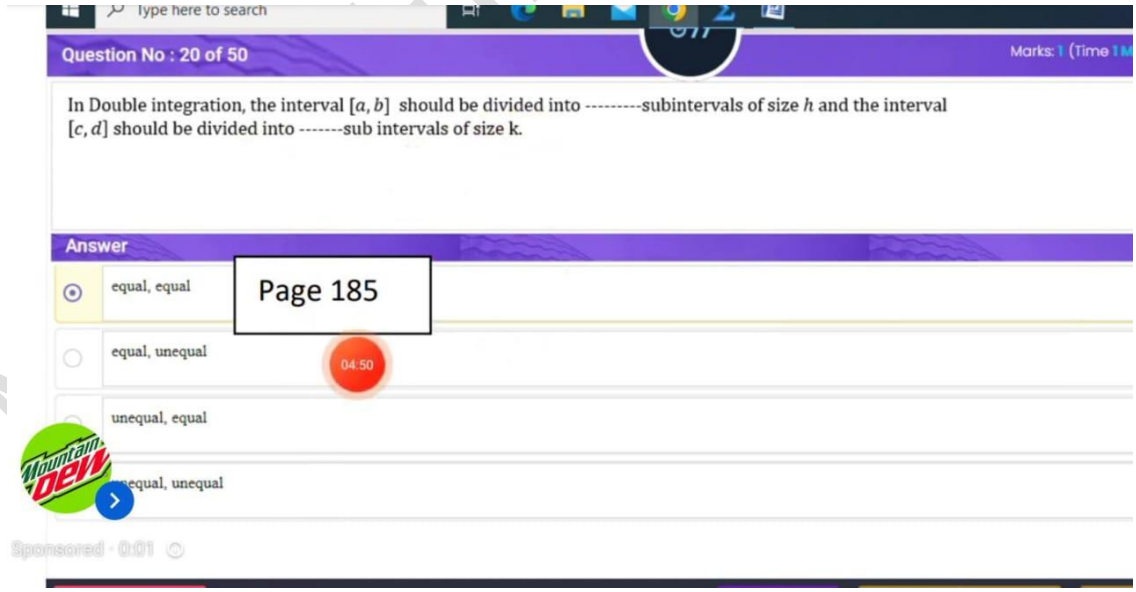

11:38 AM
In fourth order Runge-Kutta method k_4 is given by
Answer
$k_4 = hf(x_n + h, y_n + k_3)$
Page 203
$k_4 = hf(x_n + 2h, y_n + 2k_3)$
8 / 25
None of the given choices
$k_4 = hf(x_n - h, y_n - k_3)$
Finish Exam
03:48
Flag
« First
Last »

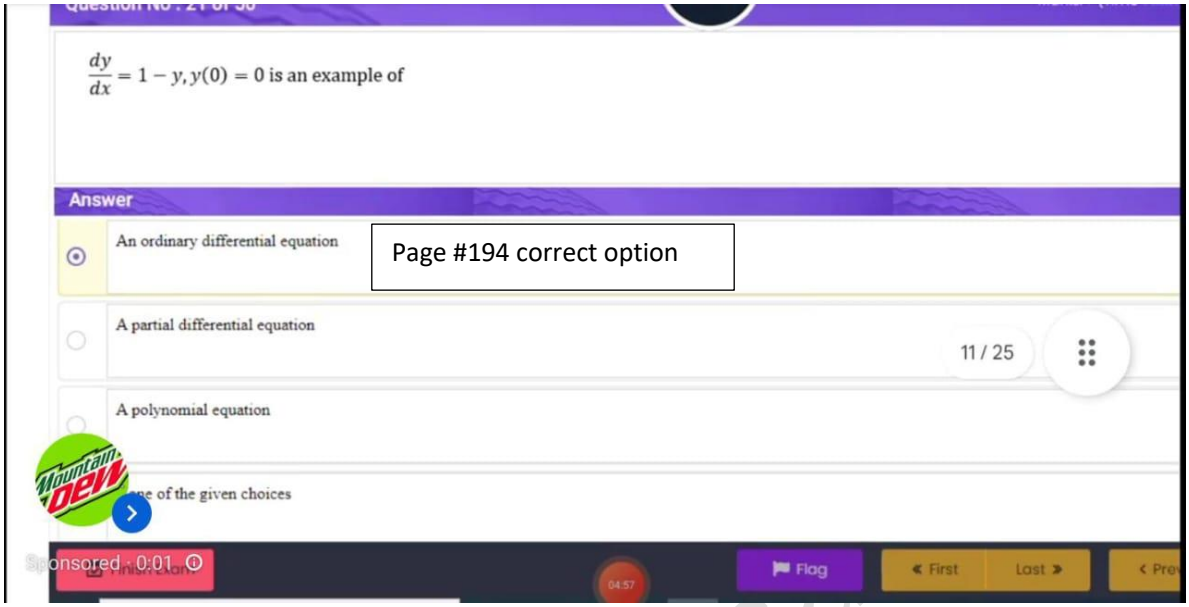

Question No : 16 of 50
Marks: 1 (Time
In integrating $\int_0^{\frac{\pi}{2}} \cos x \ dx$ by dividing the interval into four equal parts, width of the interval should be _____.
Answer
$\frac{\pi}{2}$
π
$\frac{\pi}{8}$
Correct Option
$\frac{\pi}{4}$
03:59
Finish Exam
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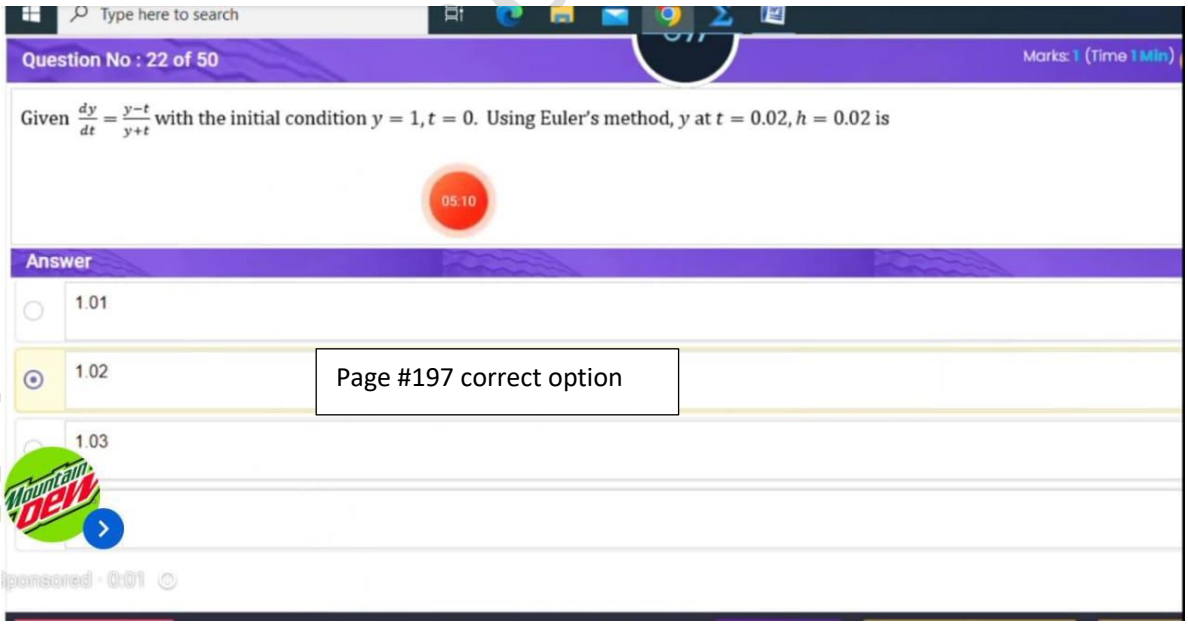

Question No : 17 of 50
Marks: 1 (Time 1 Min)
Let P be any real number and h be the step size of any interval. Then the relation between h and P for the backward difference is given by _____
Answer
$x - x_n = Ph$
$x - x_n = P$
$x + x_n = Ph$
$)h = P$
9 / 25
Sponsored · 0:01
Finish Exam
04:03
Flag
« First
Last »

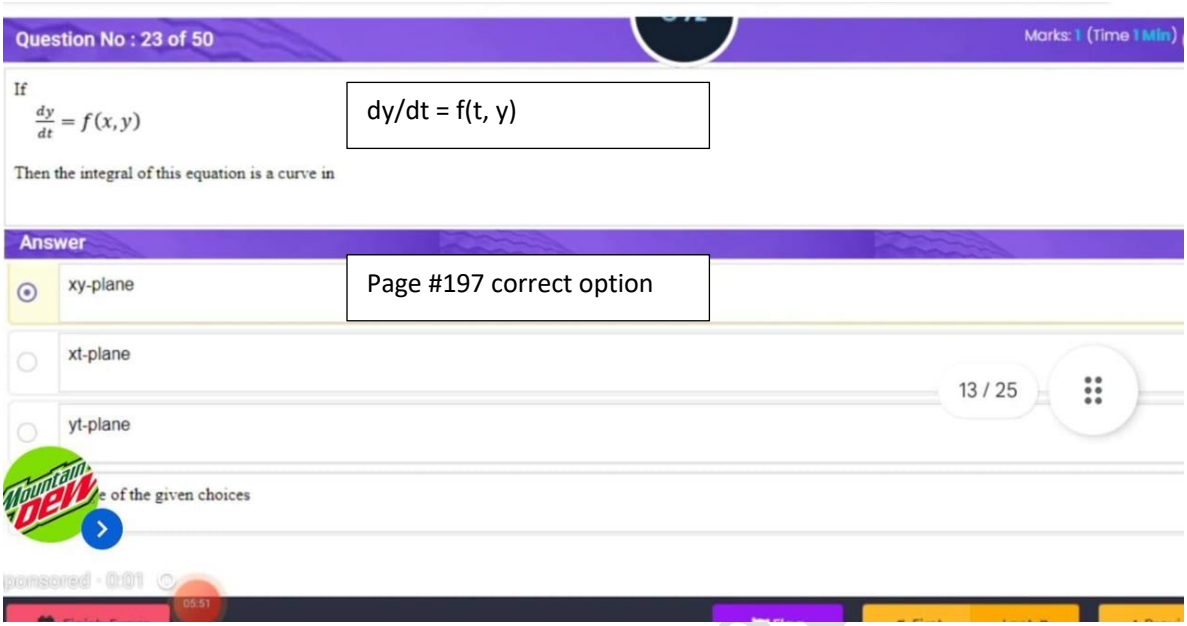
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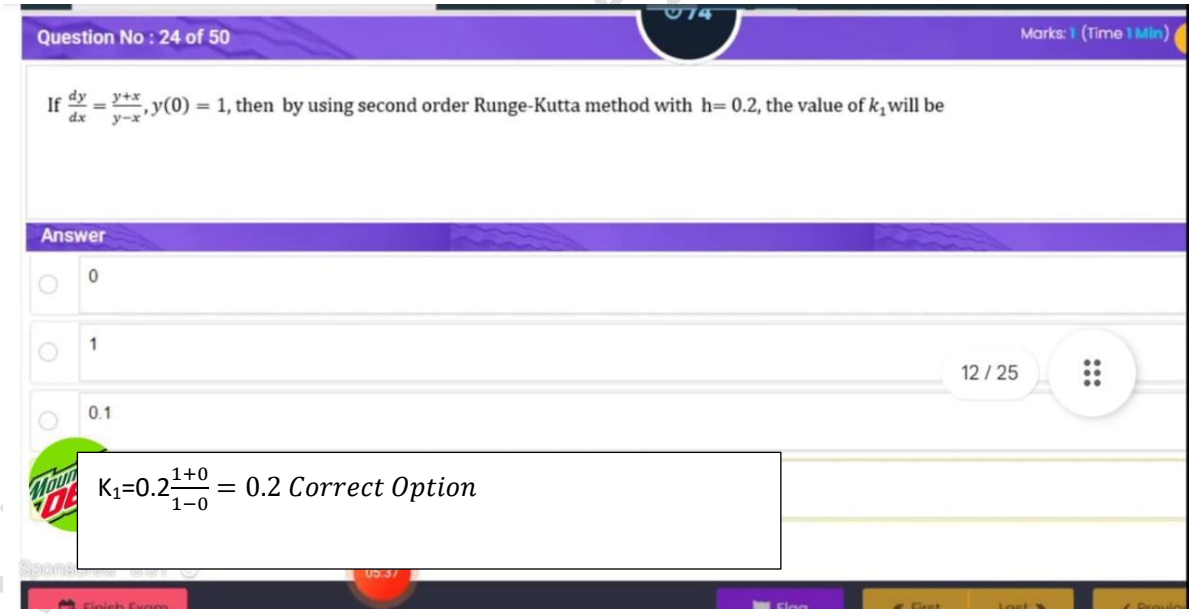

The $(n+1)^{th}$ difference of a polynomial of degree n is---.
04:39
Answer
n
0
Correct option
Constant
+1
Flag
First
Last

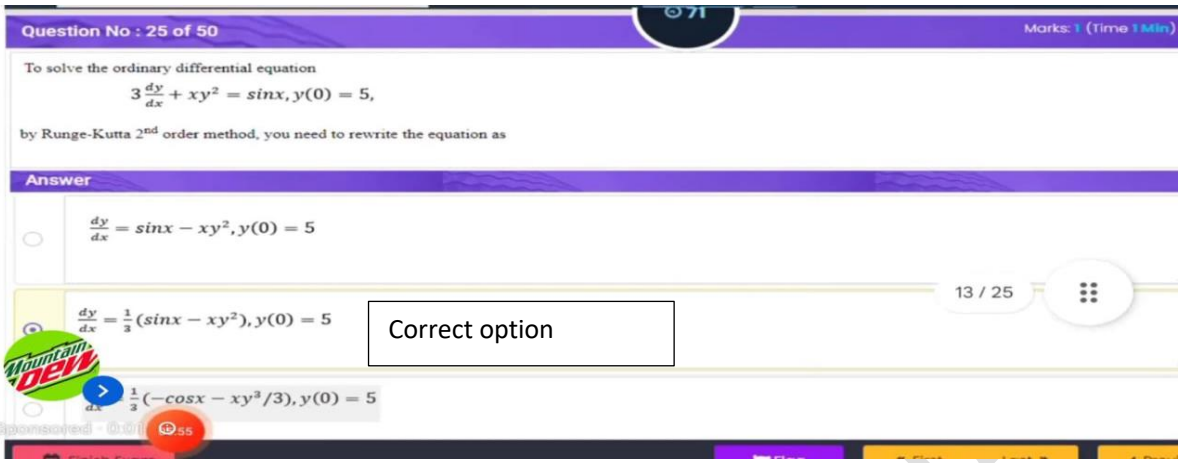

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Question No : 20 of 50
Marks: 1 (Time 1 M
In Double integration, the interval $[a, b]$ should be divided into ---------subintervals of size h and the interval $[c, d]$ should be divided into -------sub intervals of size k.
Answer
equal, equal
Page 185
equal, unequal
04:50
unequal, equal
equal, unequal

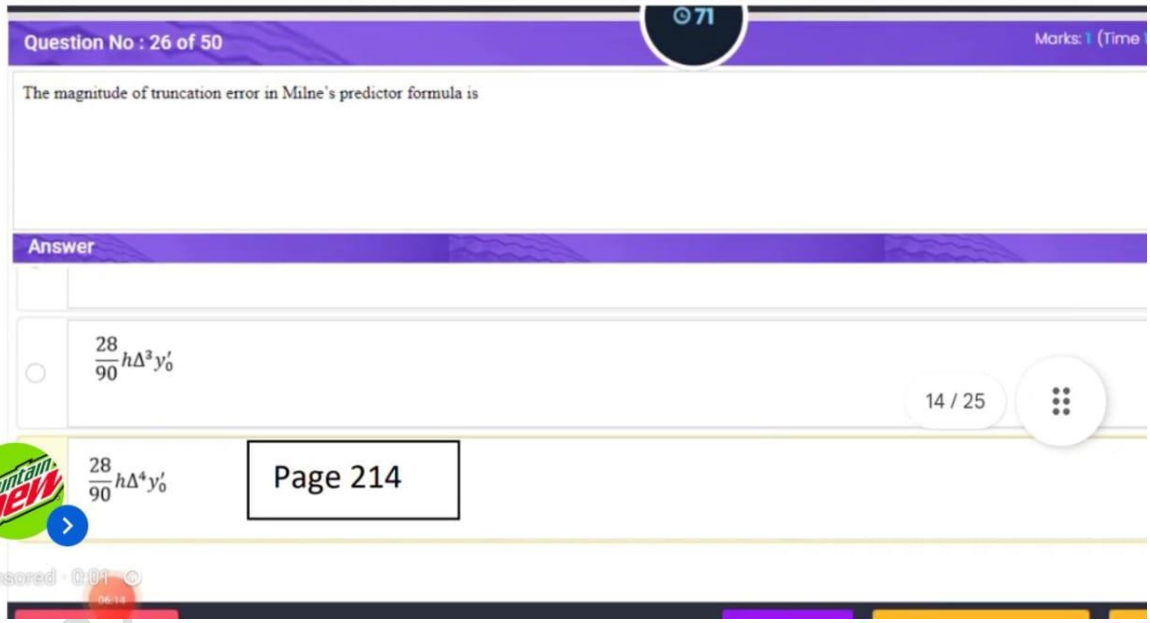

$\frac{dy}{dx} = 1 - y, y(0) = 0$ is an example of
Answer
An ordinary differential equation
Page #194 correct option
A partial differential equation
11 / 25
A polynomial equation
Flag
First
Last


Type here to search
Question No : 22 of 50
Marks: 1 (Time 1 Min)
Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1, t = 0$. Using Euler's method, y at $t = 0.02, h = 0.02$ is
05:10
Answer
1.01
1.02
Page #197 correct option
1.03


Question No : 23 of 50
Marks: 1 (Time 1 Min)
If
$\frac{dy}{dt} = f(x, y)$
Then the integral of this equation is a curve in
dy/dt = f(t, y)
Answer
xy-plane
Page #197 correct option
xt-plane
yt-plane
13 / 25
of the given choices


Question No : 24 of 50
Marks: 1 (Time 1 Min)
If $\frac{dy}{dx} = \frac{y+x}{y-x}, y(0) = 1$, then by using second order Runge-Kutta method with h= 0.2, the value of k_1 will be
Answer
0
1
0.1
12 / 25
$K_1 = 0.2\frac{1+0}{1-0} = 0.2$ Correct Option


Question No : 25 of 50
Marks: 1 (Time 1 Min)
To solve the ordinary differential equation
$3\frac{dy}{dx} + xy^2 = sinx, y(0) = 5,$
by Runge-Kutta 2nd order method, you need to rewrite the equation as
Answer
$\frac{dy}{dx} = sinx - xy^2, y(0) = 5$
13 / 25
$\frac{dy}{dx} = \frac{1}{3}(sinx - xy^2), y(0) = 5$
Correct option
$\frac{1}{3}(-cosx - xy^3/3), y(0) = 5$


Question No : 26 of 50
Marks: 1 (Time
The magnitude of truncation error in Milne's predictor formula is
Answer
$\frac{28}{90}h\Delta^3 y_0'$
14 / 25
$\frac{28}{90}h\Delta^4 y_0'$
Page 214

MK

Question No : 27 of 50

Marks: 1 (Time 1

Finding y' and y'' of the function which is tabulated below at the point

x	1.96	1.98	2.00	2.02	2.04
y	0.7825	0.7739	0.7651	0.7563	0.7473

Answer

- None of the given choices
- Forward difference operator
- Backward difference operator
- Central difference operator

14 / 25

Correct option

ponsored · 0:01

The following polynomial

$P_n(x) = y[x_0] + (x - x_0)y[x_0, x_1] + (x - x_0)(x - x_1)y[x_0, x_1, x_2] + \cdots \prod_{i=0}^{n-1}(x - x_i)y[x_0, x_1, \ldots, x_2]$

Is the --- interpolating polynomial.

Answer

- Newton's divided difference
- None of the given choices
- Newton's backward difference
- Newton's forward difference

15 / 25

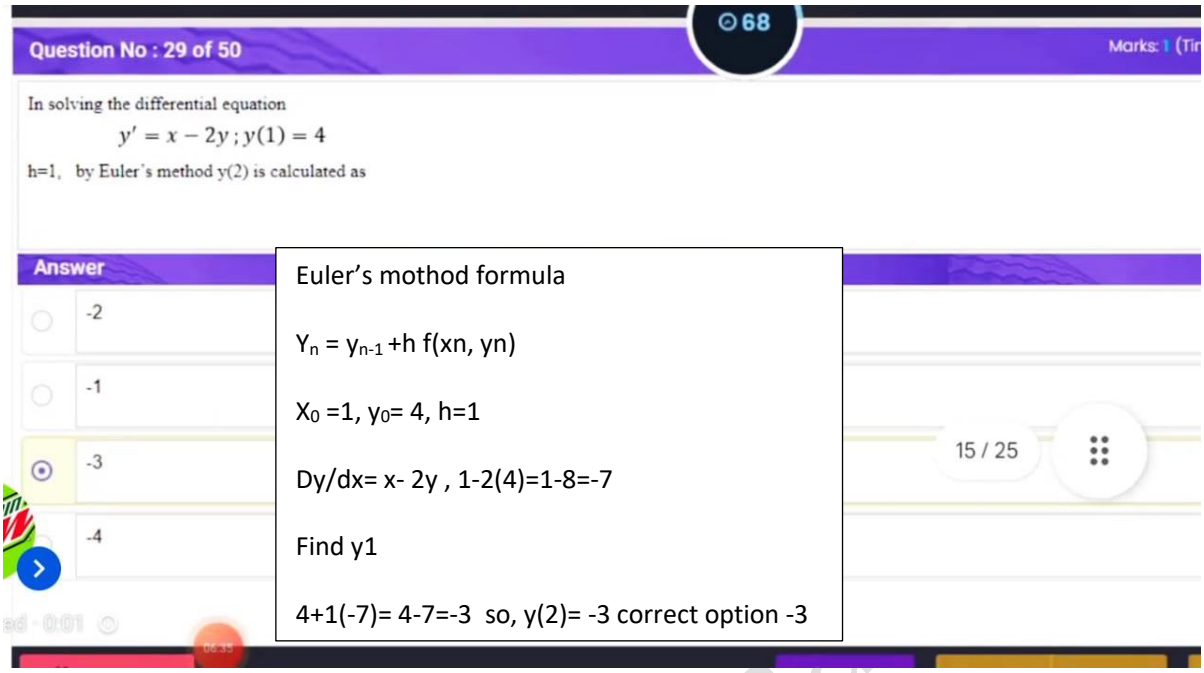
Correct option

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06:43

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« First Last »



Euler's mothod formula

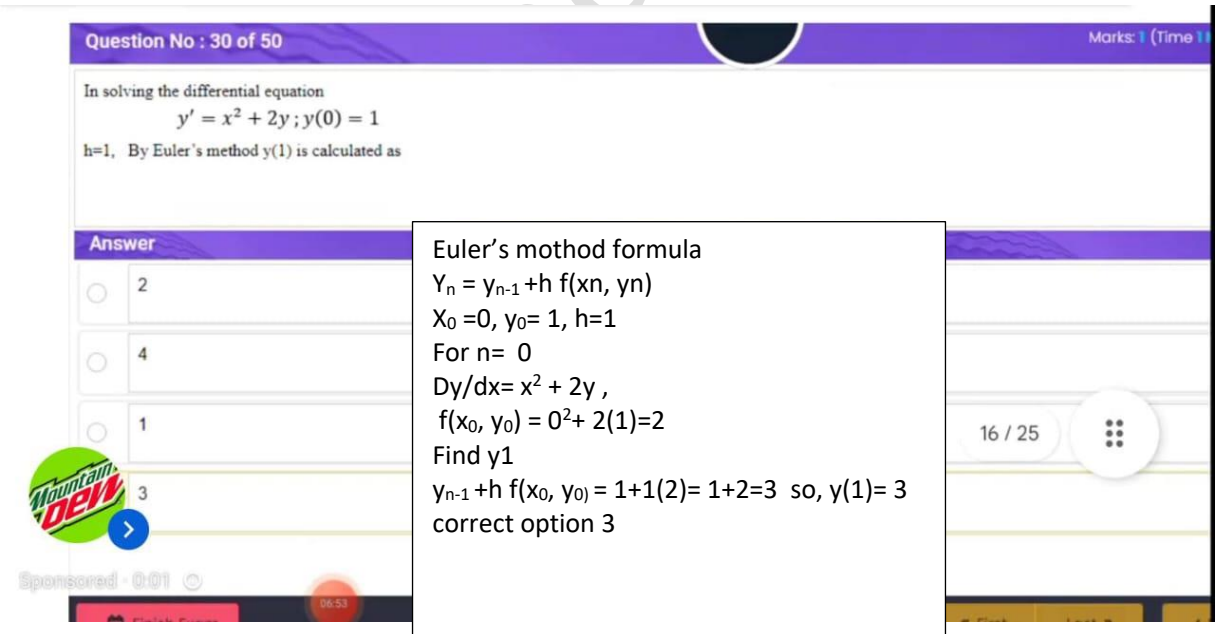
$Y_n = y_{n-1} + h\, f(xn, yn)$

$X_0 = 1$, $y_0 = 4$, $h = 1$

$Dy/dx = x - 2y$, $1 - 2(4) = 1 - 8 = -7$

Find y1

$4 + 1(-7) = 4 - 7 = -3$ so, $y(2) = -3$ correct option -3



Euler's mothod formula

$Y_n = y_{n-1} + h\, f(xn, yn)$

$X_0 = 0$, $y_0 = 1$, $h = 1$

For n= 0

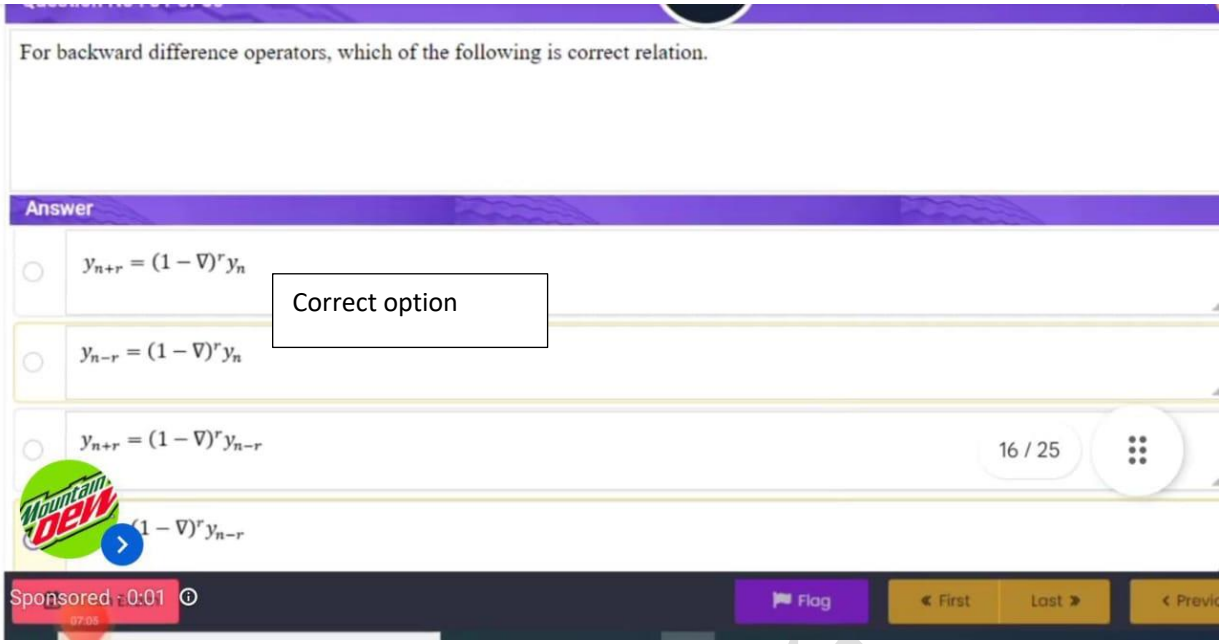
$Dy/dx = x^2 + 2y$,

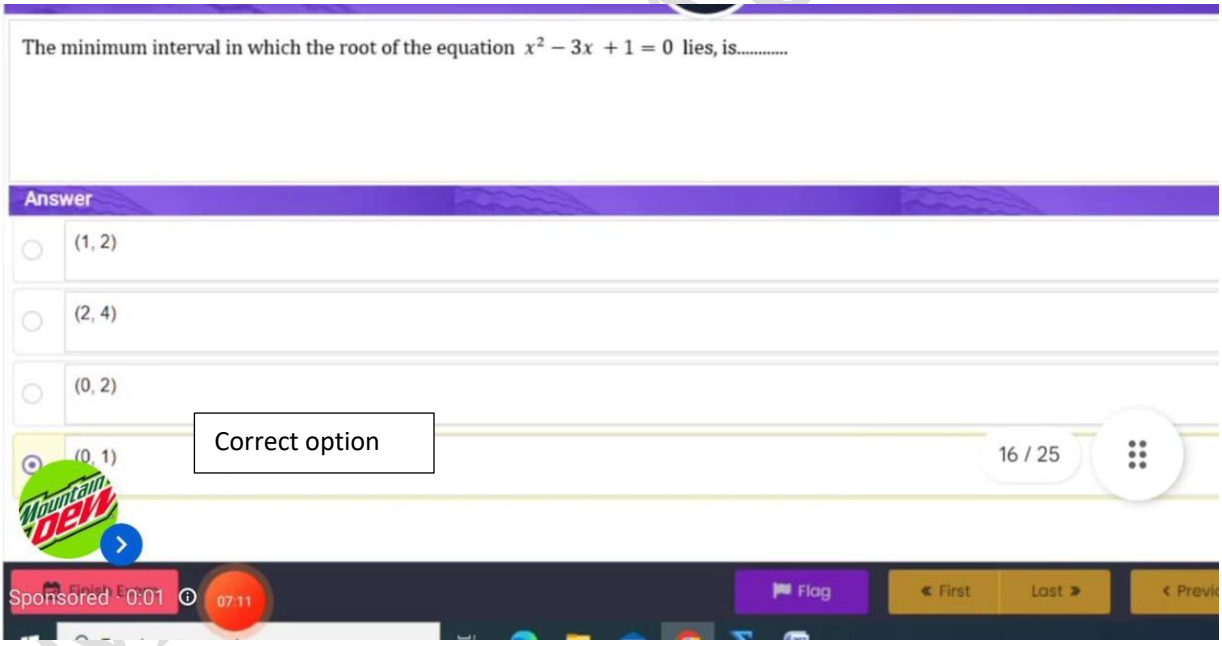
$f(x_0, y_0) = 0^2 + 2(1) = 2$

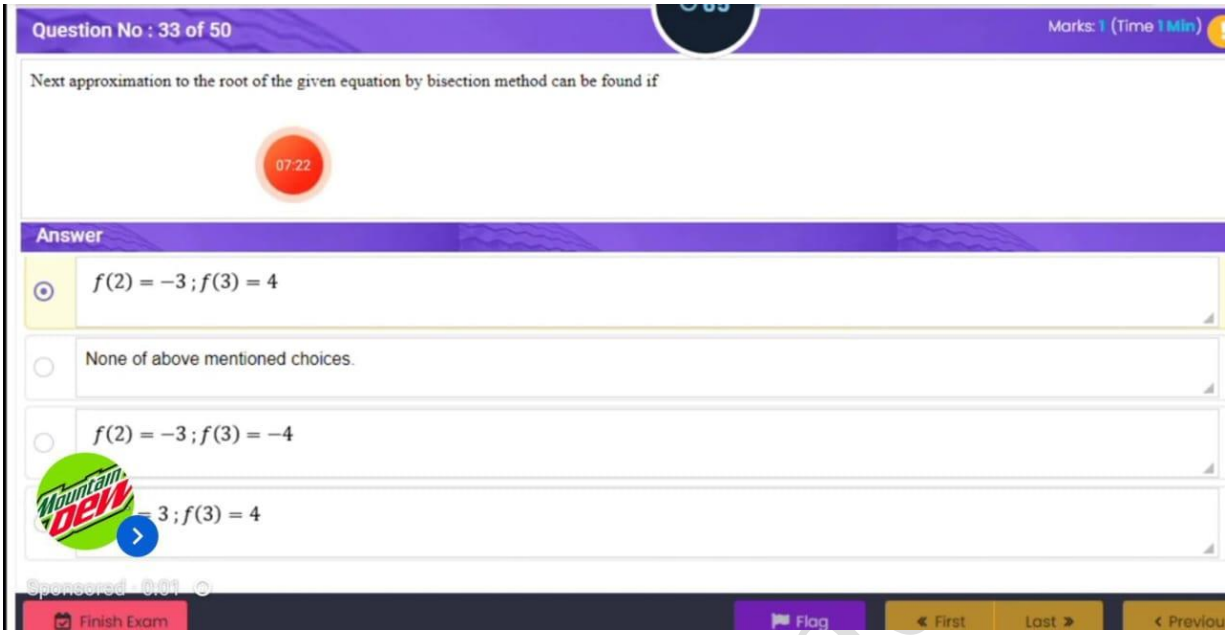
Find y1

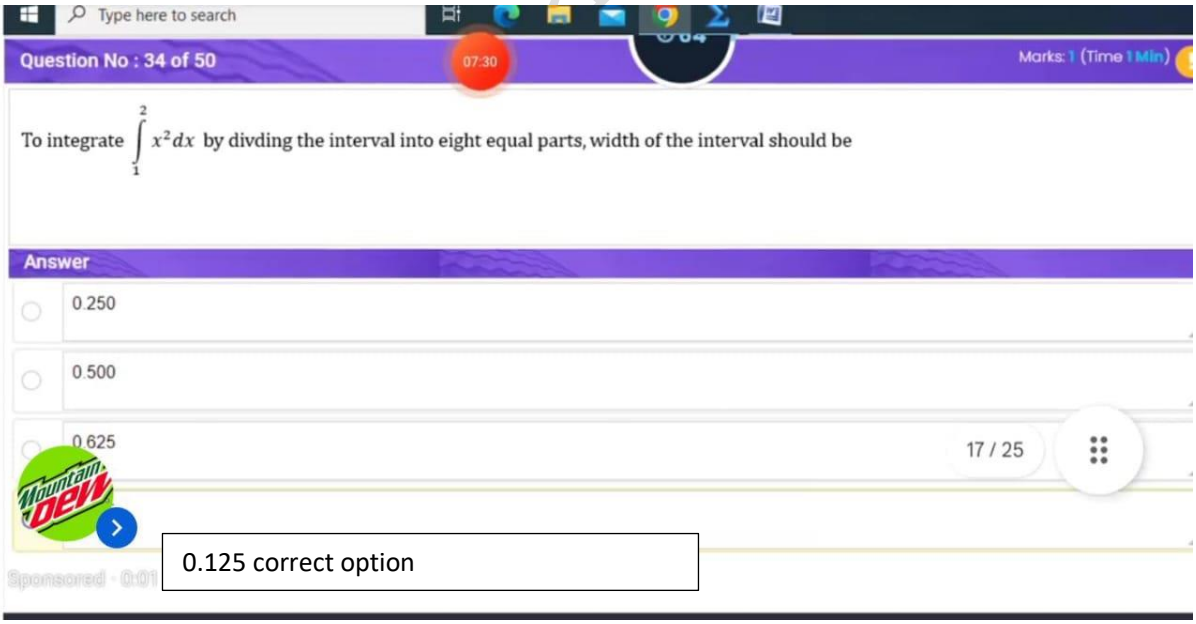
$y_{n-1} + h\, f(x_0, y_0) = 1 + 1(2) = 1 + 2 = 3$ so, $y(1) = 3$

correct option 3

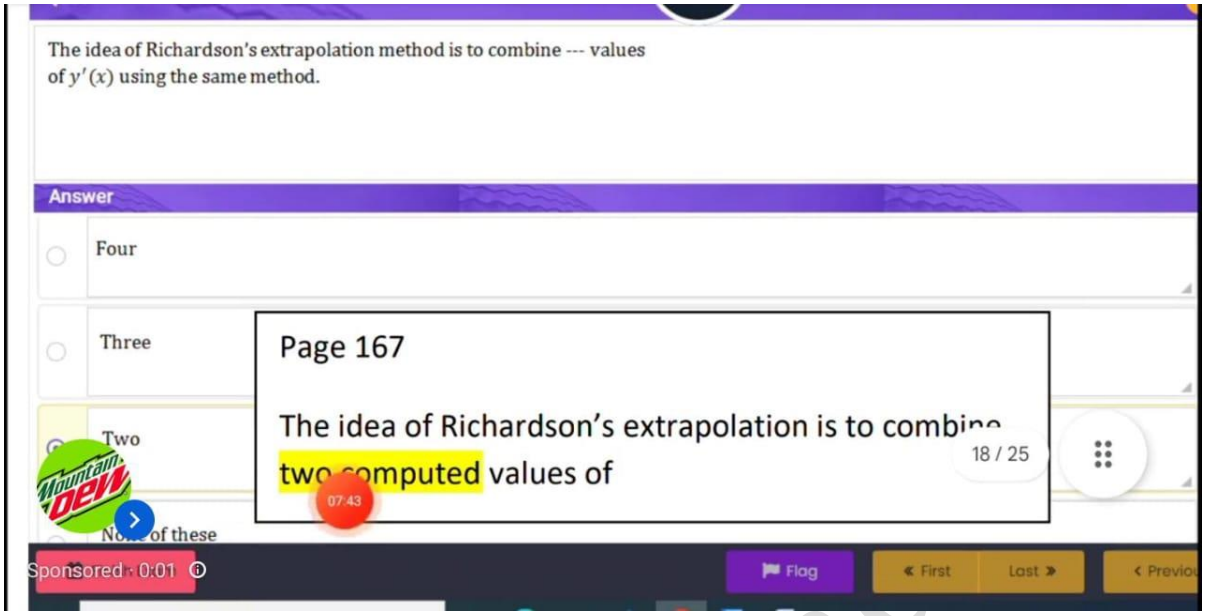

For backward difference operators, which of the following is correct relation.
Answer
$y_{n+r} = (1-\nabla)^r y_n$
Correct option
$y_{n-r} = (1-\nabla)^r y_n$
$y_{n+r} = (1-\nabla)^r y_{n-r}$
16 / 25
$(1-\nabla)^r y_{n-r}$
Sponsored 0:01
Flag
« First
Last »

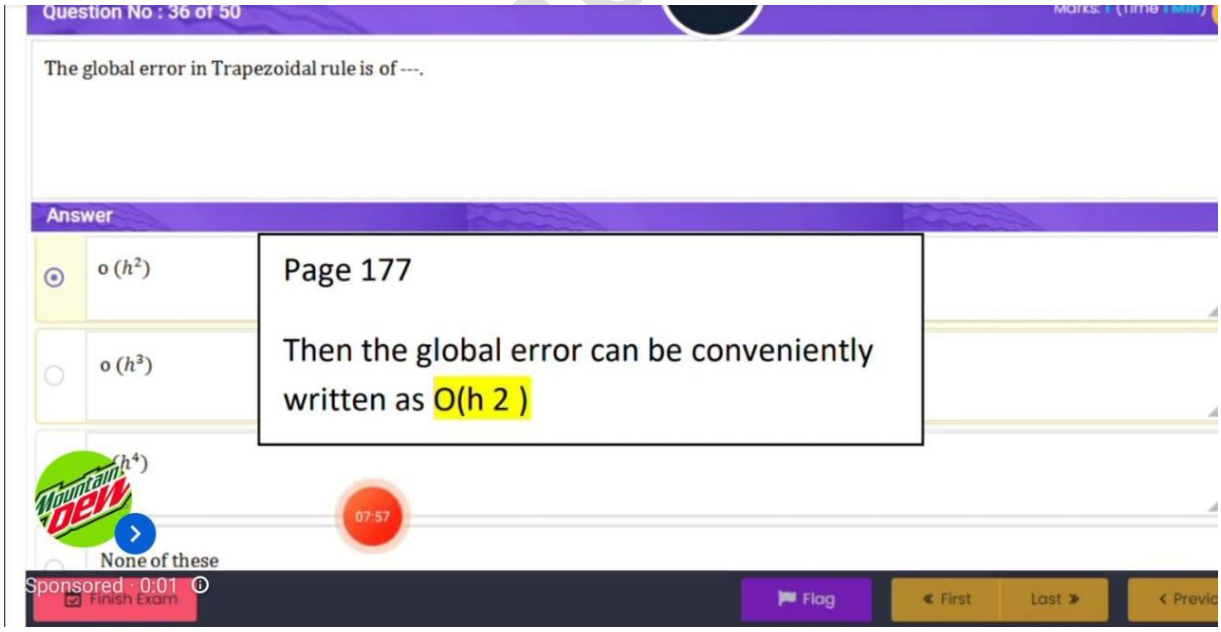

The minimum interval in which the root of the equation $x^2 - 3x + 1 = 0$ lies, is.....
Answer
(1, 2)
(2, 4)
(0, 2)
(0, 1)
Correct option
16 / 25
Sponsored 0:01
07:11
Flag
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Last »

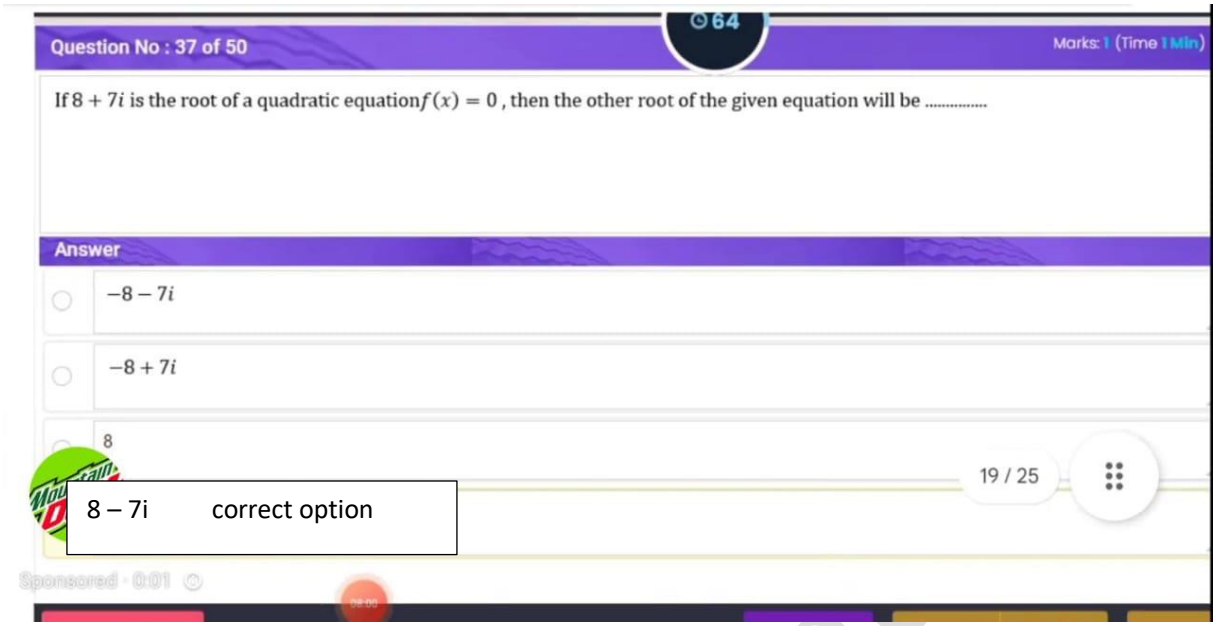




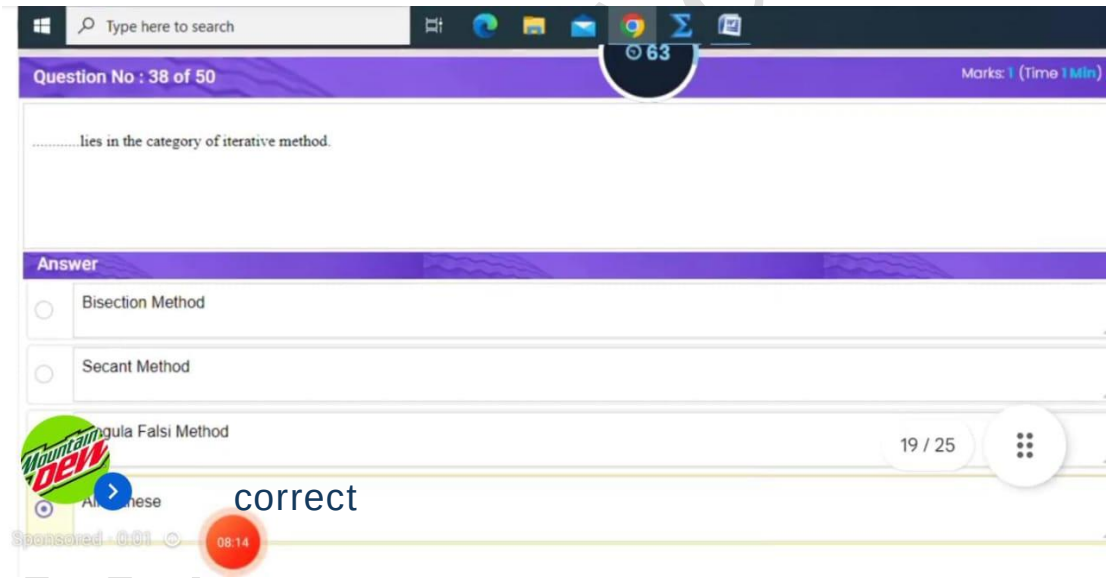
0.125 correct option


The idea of Richardson's extrapolation method is to combine --- values of $y'(x)$ using the same method.
Answer
Four
Three
Two
Page 167
The idea of Richardson's extrapolation is to combine two computed values of
18 / 25
Sponsored · 0:01
Flag
First
Last

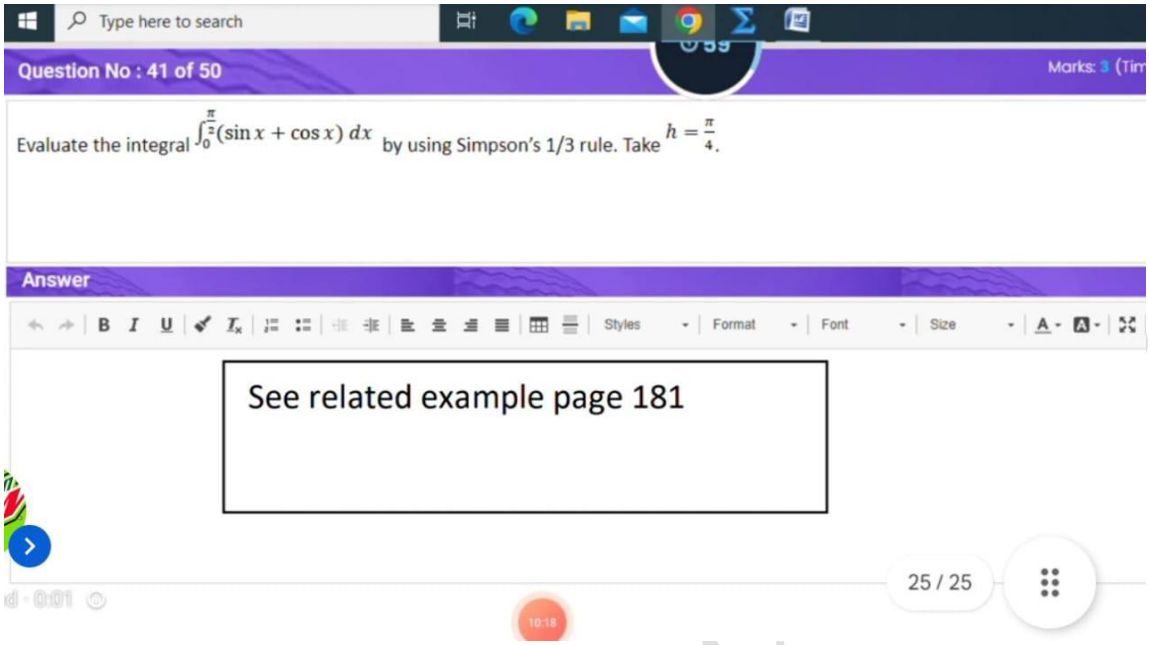

Question No : 36 of 50
The global error in Trapezoidal rule is of ---.
Answer
o (h^2)
o (h^3)
(h^4)
None of these
Page 177
Then the global error can be conveniently written as O(h 2)
Sponsored · 0:01
Finish Exam
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Last

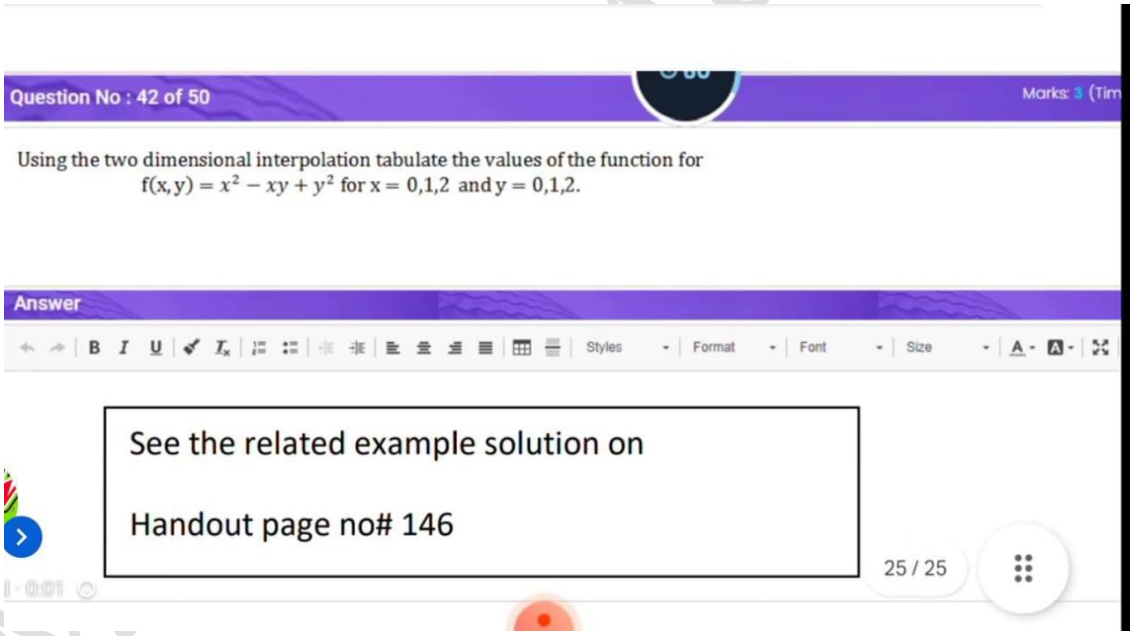


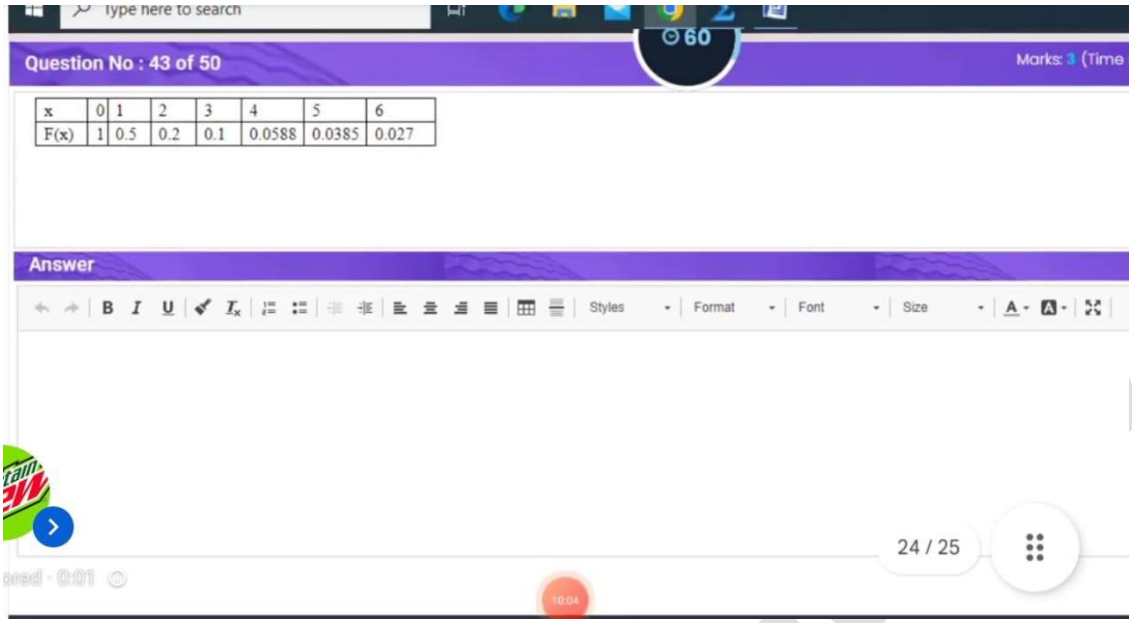
8 – 7i correct option

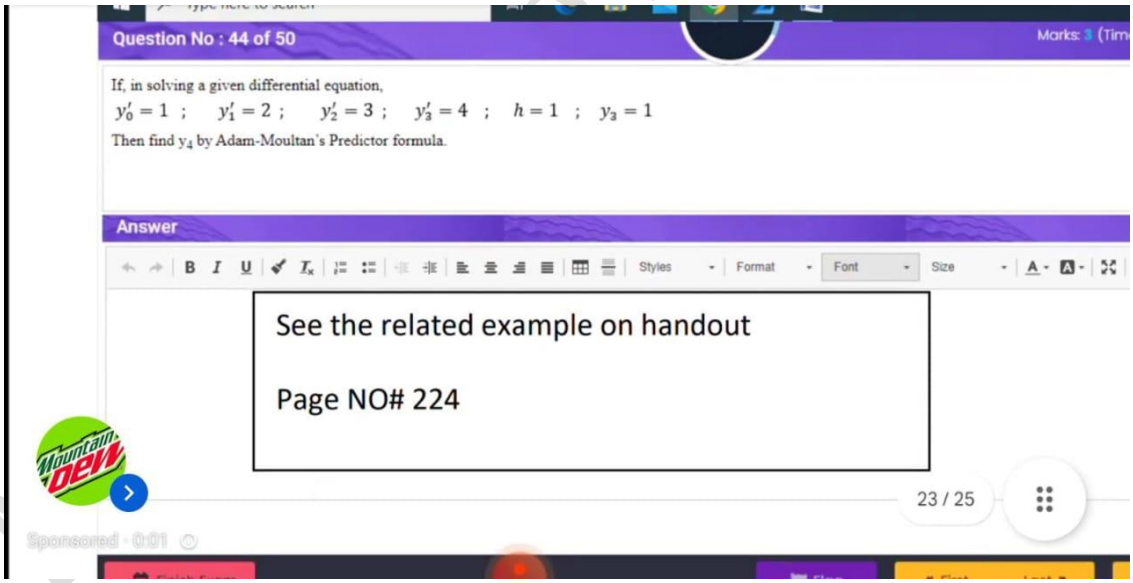


correct


Type here to search
Question No : 41 of 50
Marks: 3 (Tim
Evaluate the integral $\int_0^{\frac{\pi}{2}}(\sin x + \cos x)\,dx$ by using Simpson's 1/3 rule. Take $h = \frac{\pi}{4}$.
Answer
Styles
Format
Font
Size
See related example page 181
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Question No : 42 of 50
Marks: 3 (Tim
Using the two dimensional interpolation tabulate the values of the function for
$f(x,y) = x^2 - xy + y^2$ for x = 0,1,2 and y = 0,1,2.
Answer
Styles
Format
Font
Size
See the related example solution on
Handout page no# 146
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60
Question No : 43 of 50
Marks: 3 (Time
x 0 1 2 3 4 5 6
F(x) 1 0.5 0.2 0.1 0.0588 0.0385 0.027
Answer
Styles
Format
Font
Size
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Question No : 44 of 50
Marks: 3 (Tim
If, in solving a given differential equation,
$y_0' = 1$; $y_1' = 2$; $y_2' = 3$; $y_3' = 4$; $h = 1$; $y_3 = 1$
Then find y_4 by Adam-Moulton's Predictor formula.
Answer
Styles
Format
Font
Size
See the related example on handout
Page NO# 224
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Question No : 45 of 50

Marks: 3 (Tim

Find $F(h)$, using Richardson's extrapolation limit, while finding $y'(0.01)$ to the function

$$y = \frac{1}{x} \text{ with } h = 0.005.$$

Answer

Styles | Format | Font | Size

See the related example solved on handout

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Finish Exam | Flag | « First | Last » | ‹ Pre

Type here to search

Question No : 46 of 50

Marks: 3 (Time 6 Min

Prove that $E^{\frac{1}{2}} = \mu + \frac{\delta}{2}$

Where μ is the average operator,
δ is the central difference operator,
E is shift operator

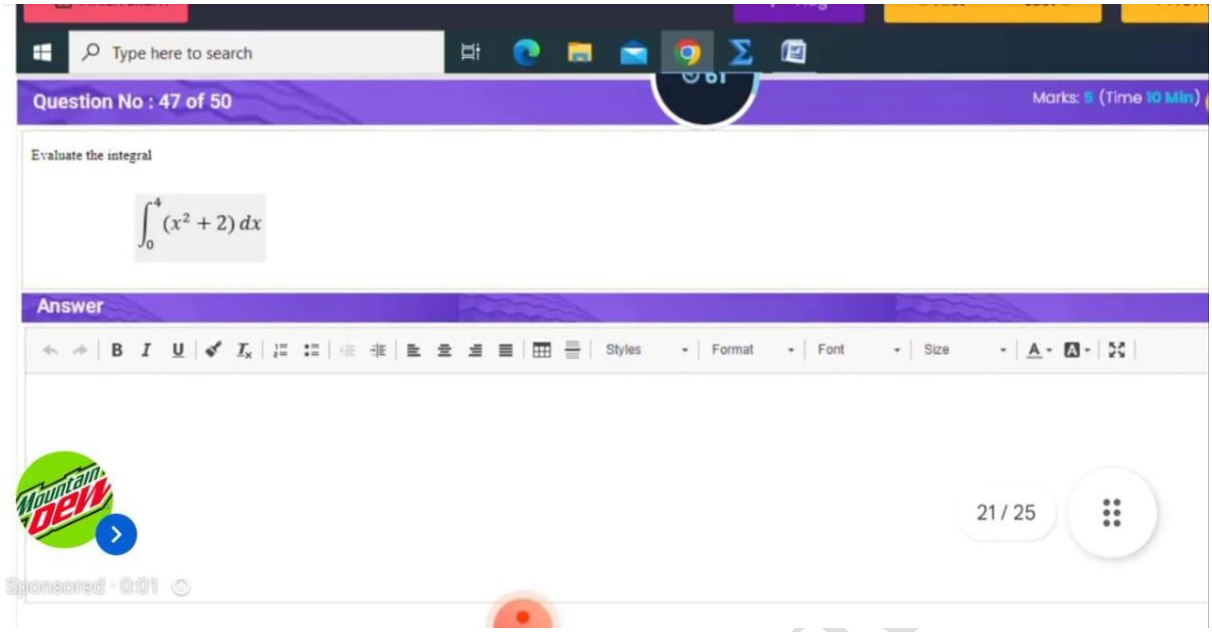
Answer

Styles | Format | Font | Size

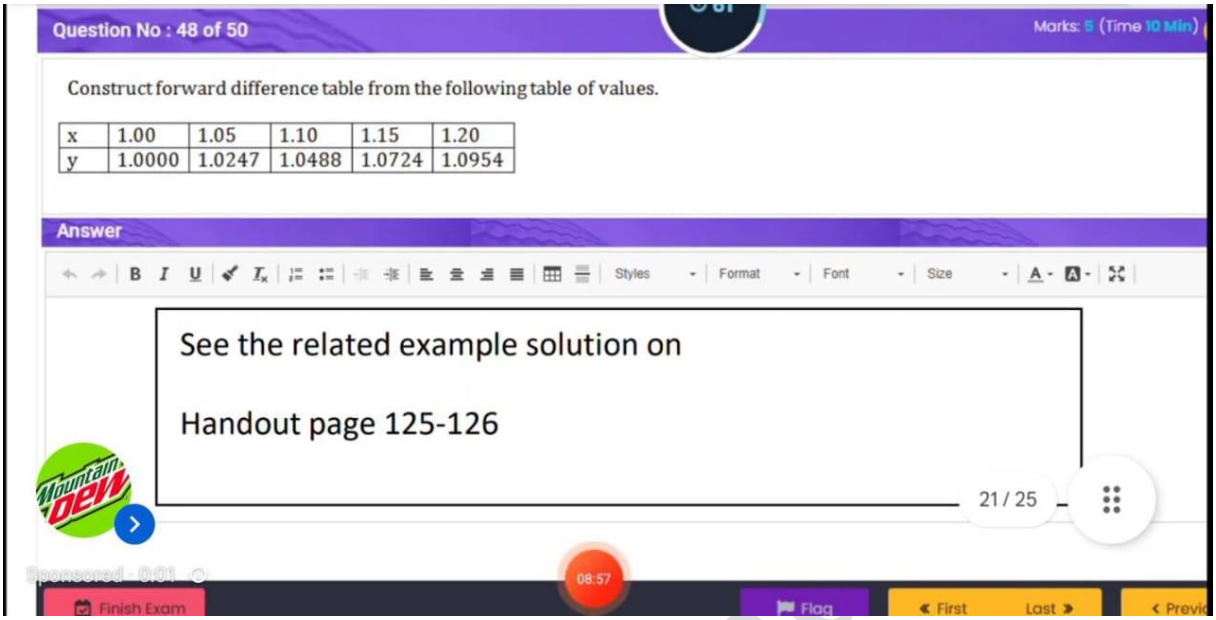
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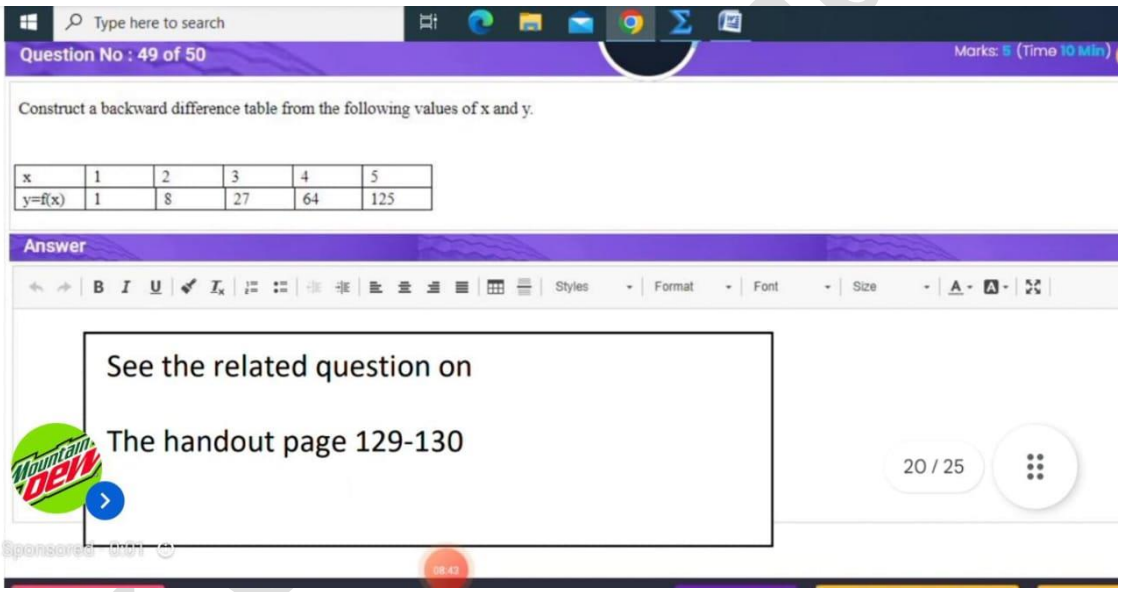
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Question No : 48 of 50
Marks: 5 (Time 10 Min)
Construct forward difference table from the following table of values.
x 1.00 1.05 1.10 1.15 1.20
y 1.0000 1.0247 1.0488 1.0724 1.0954
Answer
Styles
Format
Font
Size
See the related example solution on
Handout page 125-126
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08:57
Finish Exam
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First
Last


Type here to search
Question No : 49 of 50
Marks: 5 (Time 10 Min)
Construct a backward difference table from the following values of x and y.
x 1 2 3 4 5
y=f(x) 1 8 27 64 125
Answer
Styles
Format
Font
Size
See the related question on
The handout page 129-130
20 / 25

MK